A Low-Cost Optically Detectable Magnetic Resonance Device

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***Abstract*—Quantum sensing has become an important area of both research and technological development, yet there is a gap in accessible quantum sensing educational resources compared to a number of "freemium" computation services. To provide additional access to a wider community of learners, here we present Quorum-NV, an open-source, low-cost optically detected magnetic resonance (ODMR) device designed specifically for secondary, undergraduate, and hobbyist laboratory environments. Rather than targeting research-grade sensitivity, Quorum-NV is intended as an accessible instructional platform for introducing NV-center physics through direct experiment. The system supports foundational demonstrations such as Zeeman splitting while remaining modular enough to accommodate more advanced add-on experiments.**

I. Introduction

Quantum sensing is a mature and increasingly industrialized area of quantum technology [1]. However, educational efforts in quantum information science (QIS) have often times leaned more heavily on computation than on sensing or networking [2]. Recent work in QISE education has identified quantum sensing as an underdeveloped area within the broader quantum curriculum. Although quantum sensing is one of the major pillars of the second quantum revolution, it receives substantially less curricular attention than quantum computing [3]. More broadly, QISE course and program development benefits from modular and adaptable curricular structures, interdisciplinary collaboration, systematic evaluation, and stronger connections between academia and the quantum workforce [4]. These together motivate the development of low-cost, open, and experimentally grounded quantum sensing platforms that can be incorporated into existing courses.

In response, several groups have explored nitrogen-vacancy (NV) center diamond systems as educational platforms to introduce quantum sensing concepts [5]–[12]. Building on this emerging body of work, we present Quorum-NV, an open-source NV-center magnetometer for optically detected magnetic resonance (ODMR) experiments in the spirit of prior open quantum hardware efforts [13]. This paper provides an overview of the system design and its demonstrated educational capabilities, together with supporting design resources and an instructional build tutorial [14].

Using low-cost components, Quorum-NV enables users to observe core quantum sensing phenomena on an accessible bench-top platform. Figure 1 shows a representative wide-frequency ODMR scan centered near 2.87 GHz.

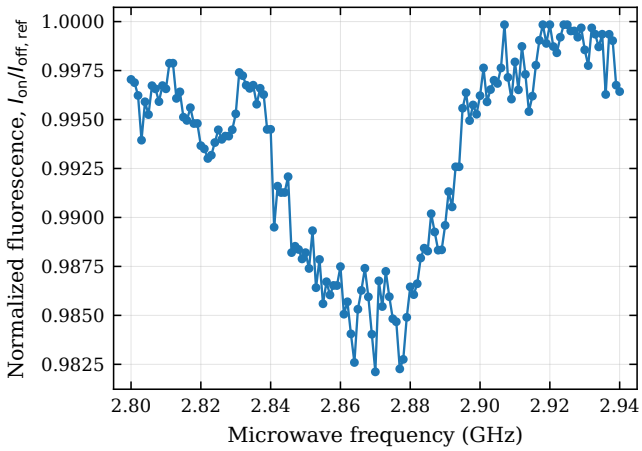


Fig. 1: Normalized fluorescence, $I_{\mathrm{on}}/I_{\mathrm{off,ref}}$, as a function of microwave frequency for the NV-center ODMR measurement acquired under an applied magnetic field of 0.5 G. A pronounced dip is observed in the vicinity of 2.87 GHz, consistent with the expected ODMR response of the NV center. Each point represents the mean of five repeated measurements acquired at that frequency. Error bars are omitted for visual clarity.

II. Related Work and Positioning

Instructional NV-center ODMR systems have developed along several complementary directions, including advanced undergraduate teaching laboratories, pedagogical magnetometry demonstrations, low-cost component studies, portable outreach devices, commercial educational systems, and open-source quantum sensing platforms. These prior efforts establish NV-center diamond systems as a practical route for introducing students to quantum sensing, while also clarifying the design space in which Quorum-NV is positioned.

An early educational reference point is the advanced undergraduate laboratory experiment of Zhang et al., which provided construction and operating instructions for optically detected

magnetic resonance (ODMR) measurements on diamond samples with high NV-center concentrations. That work demonstrated Zeeman shifts under small applied magnetic fields and emphasized the connection between core quantum concepts and an experimentally accessible solid-state spin system [15]. Sánchez Toural et al. later presented a pedagogically oriented hands-on introduction to NV^--center quantum sensing, including optical characterization, ODMR, and Zeeman-splitting measurements for magnetic-field sensing [16]. Their work is valuable as a step-by-step instructional treatment of the physics and experimental workflow, but it relies on a more conventional laboratory apparatus and is not primarily framed as a very low-cost, student-buildable open hardware kit.

Sewani et al. [5] extended the educational scope of NV-center laboratories by presenting a low-cost room-temperature teaching-lab apparatus for coherent control of an ensemble of NV^- centers in diamond. Their platform supports ODMR, spin-relaxation measurements, Rabi oscillations, and dynamical-decoupling pulse sequences, with learning outcomes that include quantum spin systems, magnetic resonance, rotating-frame physics, Bloch-sphere dynamics, and pulse-sequence development. Yang et al. [17] built on this teaching-lab apparatus to demonstrate hyperfine interactions involving the NV electronic spin and nearby ^{14}N and ^{13}C nuclear spins, using ODMR, Rabi oscillations, Ramsey-fringe experiments, and Hahn-echo sequences suitable for advanced instructional laboratories. These works show the depth of physics accessible with NV centers, but they require a more sophisticated pulsed-control laboratory environment than is needed for introductory CW-ODMR.

Yuan et al. [8] described an instructional laboratory apparatus for quantum experiments with single NV centers in diamond. In contrast to ensemble-based CW-ODMR platforms, their apparatus targets single-defect experiments and supports single-qubit initialization, rotation, and measurement, as well as more advanced investigations of electron-nuclear spin interactions. Their emphasis on students writing experimental control and data-analysis software is closely aligned with the educational goal of making the instrument transparent rather than black-boxed. However, the single-NV focus places that apparatus in a more advanced instructional-laboratory regime than the baseline Quorum-NV system, which prioritizes inexpensive ensemble CW-ODMR and introductory magnetometry.

Several efforts have focused more directly on low-cost or modular educational ODMR. Gonzalez et al. [7] presented a low-cost ODMR educational platform using nitrogen-vacancy centers in diamond. Their interactive poster documented a hands-on experimental build rather than only a conceptual proposal, including 3D-printed optical and mechanical mounts, a modular optical path, microcontroller-based control, microwave-frequency generation, optical filtering, and photodiode-based fluorescence readout. The poster materials also show RF validation using a spectrum analyzer and ODMR-style fluorescence dips near the NV-center zero-field splitting at approximately 2.87 GHz. This work is especially relevant to Quorum-NV because it demonstrates a parallel effort to make ODMR-based quantum sensing accessible through inexpensive, student-approachable hardware. In addition, correspondence with Gonzalez informed our design thinking, particularly around practical buildability, modularity, and the educational value of exposing students to the full optical, microwave, detection, and control signal chain. Stegemann et al. [6] reported a modular low-cost 3D-printed setup for NV-center experiments in diamond, emphasizing affordability, flexibility, and robustness for student and high-school use. Haverkamp et al. [9] further developed a low-cost ODMR experiment with NV centers in microdiamonds, explicitly linking technical, theoretical, and didactical considerations. These works are especially relevant to Quorum-NV because they treat affordability, modularity, and instructional accessibility as central design constraints rather than secondary considerations.

Bülau et al. [11] studied components for an inexpensive CW-ODMR NV-based magnetometer intended for educational use in schools. Their stated educational targets include fluorescence, ODMR, and Zeeman splitting, and their design emphasizes miniaturized off-the-shelf components with high reliability. Williams et al. [12] presented a portable LED-based diamond magnetometer for outreach and teaching laboratories. By replacing the typical green excitation laser with a high-power green LED, their design improves safety, reduces cost, and allows visible demonstration of green excitation and red photoluminescence. Quorum-NV instead retains a low-power green laser excitation path, which supports compact optical coupling and straightforward ODMR acquisition, but requires explicit laser-safety procedures. These approaches occupy complementary educational niches: LED excitation improves outreach safety and demonstration value, while laser excitation keeps the system closer to conventional ODMR instrumentation.

Open-source ODMR infrastructure has also emerged as an important part of the broader ecosystem. Babashah et al. [18] presented an open-source ODMR platform based on Qudi, describing both hardware considerations for ODMR experiments and software tools intended to reduce the barrier for newcomers using commercially available instrumentation. This work is not primarily a low-cost classroom kit, but it illustrates the importance of transparent and reusable control software for ODMR experiments. Carney and Kumaran [10] introduced *Uncut Gem*, a fully open-source, hackable NV-center diamond magnetometry platform. Their project emphasizes open hardware, open firmware, modularity, consumer off-the-shelf components, and democratized access to quantum sensing. Among the prior efforts cited here, this work is closest in spirit to Quorum-NV because both systems frame openness and hackability as central design requirements rather than incidental features.

Commercial educational platforms, including NV-center modules for qutools quEDU and QZabre's Quantum Edukit, provide polished turnkey access to ODMR and related experiments [19], [20]. These systems are important points of comparison because they demonstrate that NV-center quantum sensing is already recognized as an educationally valuable

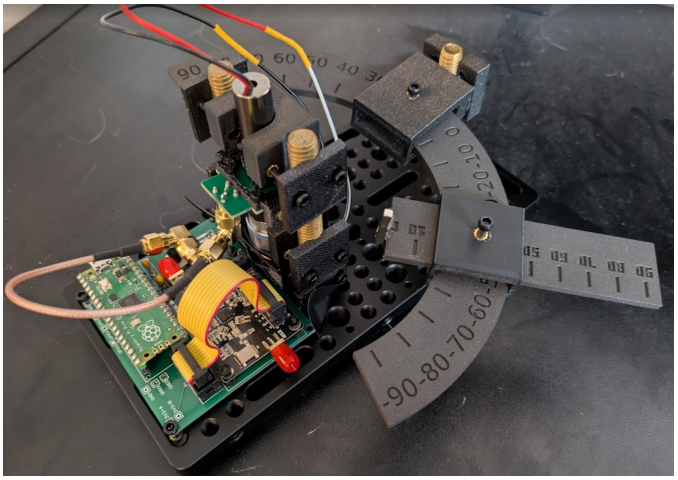


Fig. 2: A photograph of Quorum-NV with an additional magnetic scanning wand peripheral in the add-on zone.

topic. However, commercial platforms occupy a different cost, repairability, and accessibility regime than student-built open hardware. Their strength is reliability and turnkey operation; the strength of an open student-buildable platform is that the optical, microwave, detector, control, and software subsystems remain visible and modifiable.

Quorum-NV is positioned within this landscape as a low-cost, open-source, student-buildable CW-ODMR platform. It is not intended to replace commercial ODMR teaching systems, advanced single-NV laboratories, or pulsed coherent-control teaching laboratories. Instead, its contribution is to lower the entry barrier for introductory quantum sensing by exposing the full signal chain: optical excitation, microwave generation, antenna coupling, fluorescence detection, embedded control, and data analysis. By using commodity components, microcontroller-based control, a digital light sensor, and an openly documented build process, Quorum-NV emphasizes accessibility, repairability, and project-based learning rather than research-grade sensitivity or turnkey operation.

III. Approach

Our design philosophy for Quorum-NV was to create a low-cost, open-source platform that introduces NV-center physics through progressively more sophisticated experiments; while keeping the baseline system accessible to a wide range of users. Rather than optimizing for research-grade sensitivity, the design prioritizes modularity, transparency, and educational value. In this framework, the base instrument supports rapid observation of ODMR while remaining extensible to more advanced studies, including crystal-orientation dependence and nitrogen nuclear magnetic resonance (NMR) through add-on components (Figure 2).

The core measurement strategy relies on optical excitation and readout of NV-center spin states under microwave radiation. A green laser excites the diamond producing spin-dependent red photoluminescence, while a microwave field drives transitions near the NV ground-state resonance. By sweeping the microwave frequency and recording the fluorescence response, the user obtains an ODMR spectrum that links optical excitation, microwave control, and spin physics within a single bench-top experiment.

The platform is structured so that this first measurement can be obtained quickly, lowering the barrier to entry for classroom, outreach, and hobbyist use while still exposing students to authentic quantum sensing workflows. Basic ODMR can be demonstrated on the order of tens of seconds, typical scans can be completed in several minutes, and finer scans or add-on measurements may require longer acquisition times (Figure 3).

To support this educational goal, the hardware is organized as a compact optical and radio-frequency system that can be assembled, modified, and debugged by the user (Figure 4). The baseline instrument is therefore not treated as a closed black box, instead as a learning platform for optics, radio frequency (RF) electronics, spin physics, data acquisition, and device integration. The base kit is designed to remain functional at low cost while leaving electrical, optical, and software interfaces available for future extensions.

A second design element is staged capability. The first stage emphasizes robust ODMR acquisition and straightforward interpretation. Once that functionality is established, the same system can be extended with additional RF control, optical timing improvements, and higher-sensitivity detection to access more advanced phenomena. Planned add-ons include nitrogen NMR, Rabi oscillations, spin-spin interaction experiments, spin-phonon coupling, transistor-transistor logic (TTL)-capable laser control, RF switching, and avalanche-photodiode readout.

The intended workflow therefore spans multiple time scales. Short acquisitions provide immediate feedback and conceptual access, while longer scans reveal finer physical structure. Rapid measurements of roughly 30 s can demonstrate basic ODMR, scans of approximately 5 min provide more resolved spectra, and longer runs with magnetic add-ons or fine scans can extend to 45–60 min to access subtler effects such as nitrogen NMR.

IV. Components

A. Diamond Sample

The sensing element is an Adámas Nanotechnologies fluorescent NV microdiamond sample, ordered under model number MDNV700850um10mg(custom). The sample was specified as a 10 mg batch of 700–850 μm diamond particles with a nominal NV concentration of approximately 1 ppm. The large-particle microdiamond format was selected for ease of handling, optical alignment, and classroom use. Compared with nanodiamond powders or small flakes, the dry microdiamond particles can be manipulated with simple tools and positioned repeatably near the optical focus and microwave antenna.

The diamond contains an ensemble of nitrogen-vacancy centers. Under green optical excitation, the NV ensemble produces broadband red and near-infrared fluorescence, including emission associated with the negatively charged NV center.

TABLE I: Representative educational, low-cost, commercial, and open-source NV-center ODMR efforts.

Work	Primary emphasis	Representative capabilities
Zhang et al. [15]	Advanced undergraduate ODMR laboratory	ODMR, Zeeman shifts, construction and operating instructions for an experimentally accessible solid-state spin system.
Sánchez Toural et al. [16]	Pedagogical hands-on quantum sensing	Optical characterization, ODMR, Zeeman splitting, and magnetic-field sensing with NV^- centers in diamond.
Sewani et al. [5]	University quantum teaching lab	ODMR, spin relaxation, Rabi oscillations, dynamical decoupling, rotating-frame physics, and Bloch-sphere dynamics.
Yang et al. [17]	Advanced NV teaching-lab extension	Hyperfine interactions, Ramsey fringes, Hahn echo, and nuclear-spin effects in an instructional laboratory setting.
Gonzalez et al. [7]	Low-cost ODMR educational demonstration	Hands-on NV-center ODMR build with 3D-printed optical mounts, modular optical layout, microcontroller-based control, microwave-frequency generation, photodiode readout, RF validation, and observed fluorescence dips near 2.87 GHz.
Stegemann et al. [6]	Low-cost modular NV experiment	3D-printed setup, modular construction, affordability, flexibility, and robustness for student and high-school use.
Yuan et al. [8]	Single-NV instructional laboratory	Single-qubit initialization, rotation, measurement, and electron-nuclear spin experiments with single NV centers.
Haverkamp et al. [9]	Didactical low-cost ODMR with microdiamonds	ODMR, theory-experiment integration, accessible teaching sequence, and microdiamond-based quantum sensing.
Carney and Kumaran [10]	Open-source hackable quantum sensor	Consumer off-the-shelf NV magnetometry platform with open hardware, open firmware, and modular design.
Bülau et al. [11]	Inexpensive school-oriented CW-ODMR components	Fluorescence, ODMR, Zeeman splitting, miniaturized off-the-shelf components, and school-oriented reliability.
Williams et al. [12]	Portable outreach and teaching-lab magnetometer	LED-excited ODMR, improved optical safety, visible green excitation and red photoluminescence, and magnetic-object demonstrations.
qutools quEDU [19] / QZabre Edukit [20]	Commercial educational ODMR systems	Turnkey NV-center quantum sensing experiments using polished commercial instructional hardware.
Quorum-NV	Student-buildable open CW-ODMR platform	ODMR, Zeeman splitting, introductory magnetometry, microcontroller-controlled acquisition, commodity components, and open build documentation.

TABLE II: Diamond sample parameters.

Parameter	Value
Supplier	Adámas Nanotechnologies
Model number	MDNV700850um10mg(custom)
Sample format	Fluorescent NV diamond particles
Batch mass	10 mg
Particle size	700–850 μm
Nominal NV concentration	$\sim$1 ppm

TABLE III: Laser parameters.

Parameter	Value
Component type	Commercial green line diode laser module
Wavelength	520 nm
Nominal optical output power	5 mW
Beam format	Line laser
Focus	Adjustable focus
Laser architecture	Direct-emitting InGaN diode laser
Driver	Integrated constant-current driver
Housing	Compact brass module housing
Warm-up behavior	No warm-up time required
Focusing optic	40$\times$ achromatic objective
Safety treatment	Class 3R/Class IIIa

This fluorescence enables optical readout during continuous-wave optically detected magnetic resonance (CW-ODMR). In the present apparatus, a single microdiamond particle is positioned near the microwave antenna and illuminated by the 520 nm excitation laser. The resulting fluorescence is collected through the optical path and measured with a broadband light sensor.

Because the sample consists of an ensemble of NV centers distributed throughout a microdiamond particle, the measured ODMR spectrum represents an ensemble-averaged response rather than a single-defect measurement. This is advantageous for the present educational device because the increased fluorescence signal simplifies alignment and detection. However, the measured ODMR contrast, linewidth, and magnetic-field response depend on particle orientation, NV ensemble properties, optical collection efficiency, and microwave coupling. The device is therefore intended for room-temperature CW-ODMR, Zeeman-splitting measurements, and introductory magnetometry demonstrations rather than single-NV spectroscopy.

B. Laser Diode

Optical excitation is provided by a commercial 520 nm green line diode laser module with adjustable focus and nominal 5 mW optical output. The 520 nm excitation wavelength lies within the phonon-sideband absorption band of the negatively charged nitrogen-vacancy (NV^-) center, enabling optical initialization and fluorescence readout of the NV ensemble. The laser is focused onto the microdiamond using a 40$\times$ achromatic microscope objective, which also supports collection of the broadband NV fluorescence. The use of a direct diode laser module avoids the warm-up time and alignment sensitivity associated with diode-pumped solid-state green laser sources while maintaining a compact and low-cost optical excitation path.

The laser classification of the final apparatus depends on the accessible optical output power, beam geometry, focusing

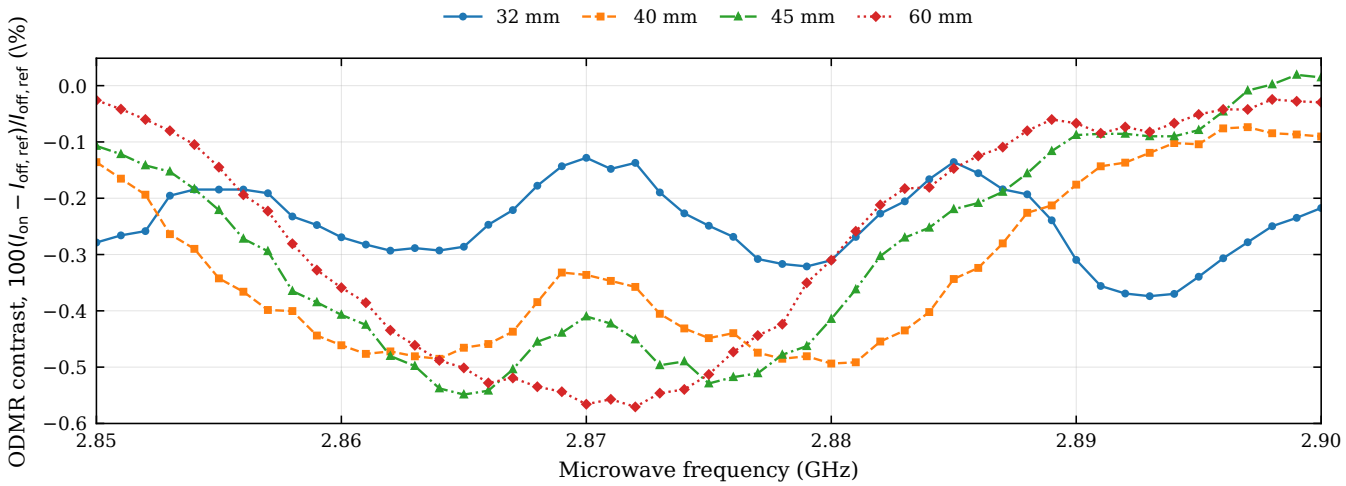


Fig. 3: ODMR contrast, $100(I_{\mathrm{on}} - I_{\mathrm{off,ref}})\,/I_{\mathrm{off,ref}}$, as a function of microwave frequency for the high-density measurements at four positions. The traces correspond to mapped positions of 32, 40, 45, and 60 mm, where the position label is defined as $60 - \mathrm{offset}$. Each curve shows the frequency-averaged data after application of a centered 5-point moving average for visual clarity. A resonance feature is observed near 2.87 GHz for all positions, with variations in depth and spectral shape across the measurement set. Note the Zeeman splitting of the peaks as the field increases.

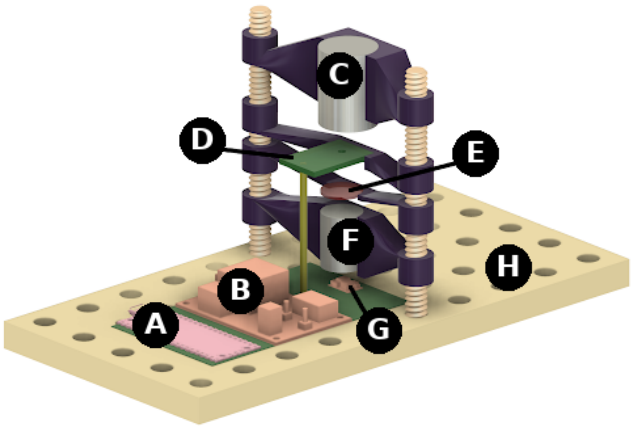


Fig. 4: Rendered schematic of the basic NV-center magnetometer construction, showing the major optical, RF, and control components. The labeled components are (A) the Raspberry Pi Pico microcontroller, (B) the radio-frequency (RF) generator, (C) the 520 nm excitation laser, (D) the RF antenna/diamond PCB, (E) the red optical filter, (F) the objective lens, (G) the optical sensor, and (H) the optical mounting board.

condition, and degree of enclosure. The laser used here is a nominal 5 mW visible laser module and should therefore not be assumed to be Class 1 or Class 2 unless the assembled device is separately evaluated and labeled as an enclosed or lower-accessible-emission laser product. OSHA guidance describes Class II visible lasers as low-power visible lasers with radiant power not above 1 mW, while Class IIIa/3R visible lasers may present a direct-viewing hazard depending on beam irradiance [21], [22]. Accordingly, the present apparatus should be treated conservatively as a Class 3R/Class IIIa laser system during assembly, alignment, and unenclosed operation.

For educational use, direct viewing of the beam and specular reflections should be avoided. The beam path should be enclosed or kept below eye level wherever practical, and alignment should be performed at the lowest usable optical power. Users should wear laser safety eyewear appropriate for 520 nm operation when alignment or other unenclosed operation is required, particularly because the microscope objective can alter the beam focus and local irradiance. Operation should occur under instructor supervision with warning labels, controlled access during operation, and safety procedures appropriate to the verified laser class of the assembled device.

C. Microwave Synthesizer and Antenna

Microwave excitation is provided by an ADF4351 fractional-N phase-locked-loop synthesizer, which provides programmable frequency control over the 35 MHz to 4.4 GHz range and sufficient frequency resolution for continuous-wave optically detected magnetic resonance (CW-ODMR) sweeps near the nitrogen-vacancy (NV) center ground-state zero-field splitting. The microwave output is coupled to the diamond using a compact loop-gap resonator adapted from the educational NV-center teaching-lab design reported by Sewani et al. [5]. The antenna includes a central optical aperture that allows the 520 nm excitation beam to be focused onto the microdiamond while the resonator provides a localized oscillating microwave magnetic field, (B_1), at the diamond.

This geometry was selected to combine microwave delivery with unobstructed optical excitation and fluorescence collection in a compact student-buildable apparatus. The de-

TABLE IV: Light sensor parameters.

Parameter	Value
Detector	TSL2591 digital light sensor
Manufacturer	ams OSRAM
Photodiode architecture	Dual silicon photodiodes
Channels	Full-spectrum and infrared-sensitive channels
Amplification	Integrated programmable transimpedance
Digitization	Integrated 16-bit analog-to-digital converter
Interface	I^2C digital interface
Programmable settings	Gain and integration time
Integration-time range	100–600 ms
Dynamic range	Approximately 600M:1

sign is also consistent with prior planar microwave antenna approaches for NV-center ODMR, including the broadband planar ring antenna demonstrated by Sasaki et al. [23]. In that work, the antenna geometry preserved optical access through a central aperture while providing microwave excitation near the NV zero-field transition. In the present device, the antenna is intended primarily for CW-ODMR, Zeeman-splitting measurements, and introductory magnetometry rather than optimized high-power pulsed spin control.

D. Fluorescence Detection

Fluorescence detection is performed using a TSL2591 digital light sensor from ams OSRAM. The TSL2591 integrates dual silicon photodiodes, programmable analog gain, a 16-bit analog-to-digital converter, and an I^2C digital interface in a compact low-cost sensor package [24]. One photodiode channel has broadband visible and near-infrared sensitivity, while the second channel is primarily infrared-sensitive. In the Quorum-NV device, the sensor is used as a broadband fluorescence detector for the red and near-infrared emission from the NV ensemble rather than as a spectrally resolved detector.

The use of an integrated digital light sensor simplifies the optical readout chain. In conventional laboratory ODMR systems, NV fluorescence is often measured using a photodiode or avalanche photodiode followed by external analog amplification and data acquisition. In the present educational device, the TSL2591 provides programmable gain, long integration times, and direct digital readout, eliminating the need for a separate low-noise transimpedance amplifier or external analog-to-digital converter. This design choice reduces cost and complexity while making the detector interface accessible to students using standard microcontroller tools.

During a continuous-wave ODMR sweep, the microcontroller sets the microwave frequency, waits for the optical signal to settle, and then reads the TSL2591 output after the selected integration interval. The measured signal is therefore a low-bandwidth, integrated fluorescence measurement rather than a time-resolved photon-counting measurement. This is appropriate for introductory CW-ODMR, Zeeman-splitting, and magnetometry demonstrations, where the relevant observable is the change in steady-state fluorescence as a function of microwave frequency. However, the sensor bandwidth and integration-time-limited readout make this configuration unsuitable for fast pulsed measurements such as time-resolved Rabi oscillations or T_1 relaxometry without substantial modification of the detection chain.

Because the TSL2591 is not calibrated as an absolute optical power meter in this apparatus, the reported ODMR spectra use relative fluorescence units derived from the raw sensor counts. For reproducibility, the gain setting, integration time, averaging procedure, and ambient-light shielding should be reported with each measurement.

TABLE V: Microcontroller parameters.

Parameter	Value
Reference controller	Raspberry Pi Pico
Microcontroller	RP2040
Logic level	3.3 V digital I/O
Microwave interface	SPI control of ADF4351 synthesizer
Optical sensor interface	I^2C readout of TSL2591 light sensor
Host interface	USB serial connection

E. Control Electronics and Software

Device control is performed using a Raspberry Pi Pico microcontroller board. The Pico was selected because it is low-cost, widely available, supported by beginner-friendly software tools, and provides the digital interfaces needed for the Quorum-NV control stack. The board is based on the RP2040 microcontroller, which includes dual Arm Cortex-M0+ cores, 264 kB of on-chip SRAM, and hardware support for common embedded interfaces including SPI, I^2C, UART, PWM, and programmable I/O [25], [26]. In the present device, the microcontroller does not generate the microwave signal directly. Instead, it programs the ADF4351 microwave synthesizer over SPI and reads the TSL2591 digital light sensor over I^2C. The ADF4351 and TSL2591 were selected in part because their digital interfaces allow the microwave sweep and fluorescence measurement to be coordinated using inexpensive embedded hardware.

During an ODMR measurement, the control software steps the microwave synthesizer through a user-defined frequency range near the NV-center ground-state splitting, waits for the microwave source and optical signal to settle, records the light-sensor response for the selected integration time, and sends the resulting frequency-dependent fluorescence data to a host computer. This division of labor keeps the timing and data acquisition requirements modest: the microcontroller handles instrument coordination, while plotting, fitting, and data analysis can be performed on a laptop or classroom computer.

The implementation is not specific to the Raspberry Pi Pico. Any microcontroller or single-board controller with suitable 3.3 V digital I/O, SPI, I^2C, and serial or USB communication could be substituted with minor software changes. This modularity is intentional, since the educational goal is to expose students to the functional signal chain of an ODMR experiment rather than to require one specific embedded platform. The Pico-based implementation provides a low-cost reference design while allowing instructors or students to replace the controller with Arduino-compatible boards,

TABLE VI: Preliminary bill of materials for the Quorum-NV base kit. Prices are approximate per-unit estimates in USD as of March 2026.

Component	Estimated Cost (USD)
ADF4351 RF synthesizer module	$26
Cables and connectors	$10
NV microdiamond	$10
520 nm laser	$15
Optical filter	$10
TSL2591 optical sensor	$7
Raspberry Pi Pico	$4
Objective lens	$12
Bolts / fasteners	$16
Custom PCBs	$8
Optical breadboard / plate	$20
3D-printed parts	20
Zip ties and miscellaneous screws	$5
Total estimated parts cost	**$163**

other RP2040 boards, ESP32-class devices, or laboratory data-acquisition hardware.

The code orchestrating this process is relatively straightforward. It is split up into python code which runs on the users PC and embedded C code which runs on the Pico. On the PC code, the user is able to set the frequency range that they would like to study, and what spacing they would like between frequency data points. The PC then sends those parameters over USB to the pico which implements them. The Pico then tells the ADF4351 RF generator which frequency to send over SMA coaxial connection to the antenna. Afterwards, it listens to the TSL2591 light sensor to get a measurement of the light. It measures the light with the RF off then on then off again and takes an average of the two off cycles which it then subtracts from the RF on cycle to get a measure of ODMR contrast. This data is then passed back over USB to the python script for saving and plotting. Finally, affordability is treated as a design requirement (see table VI. Existing educational quantum hardware often remains out of reach for many classrooms and independent learners. Quorum-NV is therefore designed as both a measurement device and an instructional framework for bringing NV-center quantum sensing into settings that would otherwise be excluded by the cost and complexity of conventional laboratory systems.

V. Educational Framing and Curriculum Integration

Quorum-NV is intended to serve two complementary educational roles. First, it can be used as a student-buildable project that introduces the practical mechanical, optical, electrical, and software skills required to construct, operate, troubleshoot, and maintain a quantum sensing device. Second, it can serve as a laboratory platform for introductory quantum information science and engineering (QISE) coursework, particularly in courses where quantum sensing is otherwise difficult to include experimentally.

Recent work has identified quantum sensing as an underrepresented area within current quantum-related curricula. Pradeep and Zwickl note that quantum sensing has received less curricular attention than quantum computing, despite being one of the major pillars of the second quantum revolution, and argue for modular quantum sensing resources that can be incorporated into existing undergraduate quantum-related courses [3]. This need is consistent with broader QISE education research emphasizing modular and adaptable curricular structures, interdisciplinary collaboration, and alignment between educational activities and workforce preparation [4]. Quorum-NV addresses this gap by providing a low-cost experimental platform that connects quantum sensing concepts to hands-on device construction and measurement.

Through building the kit, students are exposed to practical skills including circuit analysis and troubleshooting, soldering, optical alignment, microwave signal delivery, metrology, embedded programming, and data acquisition. Our team is able to assemble a complete Quorum-NV system in approximately 90 minutes. In a classroom laboratory setting, we anticipate that a team of two to four students could complete the build over two to three 90-minute class sessions, allowing additional time for instruction, debugging, safety review, and initial measurements.

Once assembled, the device can support a sequence of laboratory activities ranging from basic device characterization to introductory quantum sensing experiments. The platform is designed primarily for continuous-wave optically detected magnetic resonance (CW-ODMR), Zeeman-splitting measurements, and introductory magnetometry. These experiments allow students to connect abstract QISE concepts such as quantum states, spin-dependent fluorescence, resonance, and measurement to the behavior of a physical device.

The modularity of the platform also allows instructors to tailor the laboratory emphasis to different courses. In a modern physics or quantum mechanics course, the device can be used to connect spin physics, resonance, and magnetic-field-dependent energy shifts to an observable signal. In a QISE survey course, it can introduce quantum sensing alongside quantum computing and communication. In an electronics, instrumentation, or engineering laboratory course, the same device can emphasize microcontroller control, microwave generation, optical detection, and measurement automation. This flexibility is important because the device is not meant to replace a polished commercial ODMR teaching system. Rather, its educational value lies in making the quantum sensing signal chain visible, modifiable, and repairable.

The present implementation is optimized for CW-ODMR rather than pulsed coherent control. Experiments such as Rabi oscillations, Ramsey interferometry, spin echo, or time-resolved relaxation measurements would require additional microwave switching, timing control, and faster detection hardware. These experiments are therefore treated as future extensions rather than core capabilities of the current low-cost platform.

VI. Summary

Quorum-NV addresses a gap in quantum sensing education by providing a low-cost, open-source platform for introducing

TABLE VII: Example instructional modules supported by Quorum-NV. The modules can be used individually or combined into a multi-week laboratory sequence.

Module	Student activity	QISE and laboratory concepts	Student output
Device construction and characterization			
Build and wiring	Assemble the optical, microwave, detector, and microcontroller subsystems.	Circuit assembly, soldering, signal routing, troubleshooting, modular instrument design.	Working device checklist and subsystem test results.
Optical alignment	Align the 520 nm excitation laser to the microdiamond and maximize collected fluorescence.	Optical excitation, fluorescence collection, alignment sensitivity, photostability.	Fluorescence signal versus alignment condition.
Microwave delivery	Program the microwave source and couple the signal to the antenna near the diamond.	Frequency synthesis, microwave coupling, antenna placement, RF instrumentation.	Verified frequency sweep and antenna-position notes.
Quantum sensing measurements			
CW-ODMR	Sweep microwave frequency near the NV-center ground-state splitting and record fluorescence.	Spin-dependent fluorescence, resonance, ODMR contrast, linewidth, signal-to-noise ratio.	ODMR spectrum with fitted resonance frequency.
Zeeman splitting	Apply a static magnetic field and observe changes in the ODMR resonance structure.	Magnetic-field-dependent energy shifts, degeneracy lifting, spin Hamiltonian concepts.	Resonance splitting versus magnetic-field condition.
Magnetometry	Estimate magnetic field from the measured ODMR resonance shift or splitting.	Quantum sensing, calibration, uncertainty, sensitivity, metrology.	Magnetic-field estimate with uncertainty.
Data analysis and design iteration			
Data acquisition	Record, normalize, plot, and fit ODMR spectra using the microcontroller and host software.	Embedded programming, I^2C sensor readout, SPI device control, curve fitting.	Processed data file, plot, and fit parameters.
Noise and repeatability	Repeat measurements while varying integration time, alignment, shielding, or antenna position.	Noise sources, repeatability, averaging, design tradeoffs, experimental uncertainty.	Comparison of contrast, linewidth, and SNR across conditions.

NV-center physics through direct experiment rather than simulation alone. The system is designed to make optically detected magnetic resonance accessible in instructional, outreach, and hobbyist settings while remaining extensible enough to support more advanced experiments as users gain familiarity with the platform.

Educationally, the device brings together concepts that are often taught separately, including optical excitation, microwave control, spin-dependent fluorescence, resonance spectroscopy, and signal acquisition. By placing these ideas within a single experimental workflow, Quorum-NV offers learners a concrete entry point into quantum sensing and helps bridge the gap between theory and laboratory practice.

More broadly, this work shows that meaningful quantum education hardware can be developed with accessibility, modularity, and scalability in mind. We expect platforms of this kind to support classroom learning, curriculum development, independent projects, and future community-driven improvements in open quantum hardware [27].

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The authors disclose the use of generative AI tools, including ChatGPT-5.4, for copyediting and for troubleshooting electronic and programming issues during project development. All AI-assisted output was manually verified by the authors.

References

[1] Y. Bormuth, M. Gschwendtner, H. Soller, A. Stein, and R. L. Walsworth, "Quantum sensing can already make a difference. but where?" *Journal of Innovation Management*, vol. 12, no. 1, pp. I–XI, Sep. 2024, letter. [Online]. Available: https://journalsojs3.fe.up.pt/index.php/jim/article/view/2183-0606_012.001_L001

[2] A. Asfaw, A. Blais, K. R. Brown, J. Candelaria, C. Cantwell, L. D. Carr, J. Combes, D. M. Debroy, J. M. Donohue, S. E. Economou, E. Edwards, M. F. J. Fox, S. M. Girvin, A. Ho, H. M. Hurst, Z. Jacob, B. R. Johnson, E. Johnston-Halperin, R. Joynt, E. Kapit, J. Klein-Seetharaman, M. Laforest, H. J. Lewandowski, T. W. Lynn, C. R. H. McRae, C. Merzbacher, S. Michalakis, P. Narang, W. D. Oliver, J. Palsberg, D. P. Pappas, M. G. Raymer, D. J. Reilly, M. Saffman, T. A. Searles, J. H. Shapiro, and C. Singh, "Building a quantum engineering undergraduate program," *IEEE Transactions on Education*, vol. 65, no. 2, pp. 220–242, May 2022.

[3] N. Pradeep and B. M. Zwickl, "Analysis of quantum-related courses and textbooks for potential integration of quantum sensing," 2026.

[4] S. El-Adawy, A. R. Piña, B. M. Zwickl, and H. J. Lewandowski, "Insights from educators on building a more cohesive quantum information science and engineering education ecosystem," *Physical Review Physics Education Research*, vol. 21, p. 020144, 2025.

[5] V. K. Sewani, H. H. Vallabhapurapu, Y. Yang, H. R. Firgau, C. Adambukulam, B. C. Johnson, J. J. Pla, and A. Laucht, "Coherent control of NV^- centers in diamond in a quantum teaching lab," *American Journal of Physics*, vol. 88, no. 12, pp. 1156–1169, 2020.

[6] J. Stegemann, M. Peters, L. Horsthemke, N. Langels, P. Glösekötter, S. Heusler, and M. Gregor, "Modular low-cost 3d printed setup for experiments with NV centers in diamond," *European Journal of Physics*, vol. 44, no. 3, p. 035402, 2023.

[7] J. L. Gonzalez, D. Chapman, R. Behunin, and T. Thornton. (2023) Quantum sensing: Low cost optically detected magnetic resonance experiments using nitrogen vacancy centers in diamonds. Interactive poster. [Online]. Available: https://aas242-aas.ipostersessions.com/?s=6C-DC-81-18-82-0B-7E-A6-A0-89-E4-3A-D5-34-E3-9D

[8] Z. Yuan, S. Mukherjee, J. D. Thompson, N. P. de Leon, A. Gardill, and S. Kolkowitz, "An instructional lab apparatus for quantum experiments with single nitrogen-vacancy centers in diamond," *American Journal of Physics*, vol. 92, no. 11, pp. 892–900, 11 2024. [Online]. Available: https://doi.org/10.1119/5.0216511

[9] N. Haverkamp, A. Pusch, M. Gregor, and S. Heusler, "Low-cost odmr experiments with nitrogen-vacancy centers in diamonds: a didactical approach to theory and experiment," *EPJ Quantum Technology*, vol. 12, no. 1, 2025, published 21 February 2025. [Online]. Available: https://link.springer.com/article/10.1140/epjqt/s40507-025-00326-5

[10] M. Carney and V. Kumaran, "Uncut gem - an open-source hackable quantum sensor," in *Quantum Engineering Sciences and Technologies for Industry and Services*, ser. Communications in Computer and Information Science, F. Barbaresco and F. Gerin, Eds. Cham: Springer, 2026, vol. 2743, qUEST-IS 2025.

[11] A. Bülau, D. Walter, and K.-P. Fritz, "Components for an inexpensive cw-odmr nv-based magnetometer," *Magnetism*, vol. 5, no. 3, 2025. [Online]. Available: https://doi.org/10.3390/magnetism5030018

[12] H. Williams, A. Newman, S. Graham, C. Stephen, and G. Morley, "A portable led-based diamond magnetometer for outreach and teaching labs," *American Journal of Physics*, vol. 94, no. 6, pp. 489–493, 06 2026. [Online]. Available: https://doi.org/10.1119/5.0295777

[13] N. Shammah, A. Saha Roy, C. G. Almudever, S. Bourdeauducq, A. Butko, G. Cancelo, S. M. Clark, J. Heinsoo, L. Henriet, G. Huang, C. Jurczak, J. Kotilahti, A. Landra, R. LaRose, A. Mari, K. Nowrouzi, C. Ockeloen-Korppi, G. Prawiroatmodjo, I. Siddiqi, and W. J. Zeng, "Open hardware solutions in quantum technology," *APL Quantum*, vol. 1, no. 1, pp. 011 501–011 501–17, 2024.

[14] Viti, Ivan, "Diy quantum nv-center magneto-meter," YouTube video, 2026. [Online]. Available: https://www.youtube.com/watch?v=-hbqtUVD_Ps

[15] H. Zhang, C. Belvin, W. Li, J. Wang, J. Wainwright, R. Berg, and J. Bridger, "Little bits of diamond: Optically detected magnetic resonance of nitrogen-vacancy centers," *American Journal of Physics*, vol. 86, no. 3, pp. 225–236, 2018.

[16] J. L. Sánchez Toural, V. Marzoa, R. Bernardo-Gavito, J. L. Pau, and D. Granados, "Hands-on quantum sensing with NV^- centers in diamonds," *C*, vol. 9, no. 1, p. 16, 2023.

[17] Y. Yang, H. H. Vallabhapurapu, V. K. Sewani, M. Isarov, H. R. Firgau, C. Adambukulam, B. C. Johnson, J. J. Pla, and A. Laucht, "Observing hyperfine interactions of NV^- centers in diamond in an advanced quantum teaching lab," *American Journal of Physics*, vol. 90, no. 7, pp. 550–560, 2022.

[18] H. Babashah, H. Shirzad, E. Losero, V. Goblot, C. Galland, and M. Chipaux, "Optically detected magnetic resonance with an open source platform," *SciPost Physics Core*, vol. 6, p. 065, 2023.

[19] qutools GmbH, *quEDU Education Kit: quADD-NV Experiment Board, Manual V1.2*, qutools GmbH, München, Germany, 2024, manual V1.2. [Online]. Available: https://www.qutools.com

[20] QZabre AG, "Quantum Edukit: Hands-on NV Center Teaching Kit for University Physics Courses," https://www.qzabre.com/quantum-edukit, accessed 2026-07-22.

[21] Occupational Safety and Health Administration, "Guidelines for laser safety and hazard assessment," https://www.osha.gov/enforcement/directives/std-01-05-001, 1991, accessed 2026-07-22.

[22] ——, "Laser hazards," https://www.osha.gov/laser-hazards/hazards, accessed 2026-07-22.

[23] K. Sasaki, Y. Monnai, S. Saijo, R. Fujita, H. Watanabe, J. Ishi-Hayase, K. M. Itoh, and E. Abe, "Broadband, large-area microwave antenna for optically detected magnetic resonance of nitrogen-vacancy centers in diamond," *Review of Scientific Instruments*, vol. 87, no. 5, p. 053904, 05 2016. [Online]. Available: https://doi.org/10.1063/1.4952418

[24] ams OSRAM, "TSL2591 Ambient Light Sensor Datasheet," https://look.ams-osram.com/m/c901de8e97608f8/original/TSL2591-DS000338%20Datasheet, accessed 2026-07-22.

[25] Raspberry Pi Ltd, "Raspberry Pi Pico," https://www.raspberrypi.com/products/raspberry-pi-pico/, accessed 2026-07-22.

[26] ——, "RP2040 Specifications," https://www.raspberrypi.com/products/rp2040/specifications/, accessed 2026-07-22.

[27] S. A. Metwalli, A. Iliat, S. Thomas, S. Nair, Z. A. Chase, and R. R. Ceballos, "The quantum education ecosystem: A review of global initiatives, methods, and challenges," arXiv preprint, Apr. 2026. [Online]. Available: https://arxiv.org/abs/2604.06293